Emergent Branch Selection and Finite Measurement Thickness in a Quantum Simulation Framework

Saku Hämäläinen
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We investigate measurement-like behavior in a numerical quantum simulation framework equipped with an emergent detector model but no explicit collapse term. In baseline Schrödinger evolution, the wave field spreads through multiple available paths and generates the expected interference structure at the screen. However, the detector outputs are not observed as a passive sampling of the continuous Born density. Instead, detection events cluster into a small number of discrete branches.

This behavior suggests that branch selection can arise from detector dynamics, coarse graining, and propagation geometry, even without introducing an explicit nonlinear collapse operator into the wave evolution itself. The results further indicate that the selection process is not pointlike in time. Rather, it appears to develop over a finite spatiotemporal region, which we describe as a finite measurement thickness.

We propose a geometric test configuration consisting of a wide slit followed by a downstream double slit with variable separation distance. In this setup, the distance between structures controls a transition between near-field, intermediate, and far-field regimes. This provides a direct numerical probe of whether branch selection is gradual and geometry dependent.

The framework does not claim to replace standard quantum mechanics. Instead, it offers an exploratory computational model for studying how discrete detector outcomes may emerge from continuous wave dynamics through a finite, structured, measurement process.

INTRODUCTION

Quantum mechanics describes the evolution of a system through the Schrödinger equation

$$i\hbar\partial_t\psi = H\psi. \tag{1}$$

This unitary dynamics successfully predicts interference, diffraction, and a wide range of experimentally verified phenomena. However, the relation between the continuous wave description and the appearance of definite detection outcomes remains conceptually nontrivial.

In standard textbook treatments, a measurement outcome is associated with the Born rule, according to which the probability density is

$$\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2. \tag{2}$$

This rule predicts outcome frequencies, but it does not by itself specify a mechanism by which a single event becomes definite in an individual run.

The purpose of the present work is not to propose a new fundamental law of quantum dynamics, but to explore a computational model in which discrete measurement outcomes emerge from ordinary wave propagation together with a structured detector model. The central question is not only *which* distribution is obtained, but *how* branch selection develops.

Our numerical observations suggest two main points. First, detection outputs are not always a direct passive sampling of the continuous Born density. Instead, they often organize into a small number of discrete branches. Second, this selection appears to occur gradually over a finite spatiotemporal region rather than instantaneously at a single point. We refer to this as *finite measurement thickness*.

These observations motivate a new class of numerical experiments aimed at probing where, when, and how branch selection occurs.

BASELINE WAVE EVOLUTION

We begin from ordinary Schrödinger evolution in a two-dimensional geometry. An initial Gaussian packet is propagated toward a barrier structure using a split-operator method. In the absence of any detector-specific modification, the evolution produces the familiar diffraction and interference patterns.

Forward evolution from an initial state ψ_0 is written as

$$\psi(\mathbf{r}, t) = U(t, t_0)\psi_0(\mathbf{r}). \tag{3}$$

The corresponding Born density is

$$\rho(\mathbf{r}, t) = |\psi(\mathbf{r}, t)|^2. \tag{4}$$

For a double-slit geometry, ρ typically develops an interference pattern downstream of the barrier. This continuous field structure serves as the reference against which detector outputs are compared.

EMERGENT DETECTOR MODEL

The detector model considered here is intentionally exploratory. It does not insert an explicit collapse term into the Schrödinger equation. Instead, discrete outcomes are

extracted from the evolving field by a structured detection process involving finite spatial resolution, local maxima selection, and branch-like clustering of clicks.

At the most general level, the detector is represented by a map

$$\mathcal{D} : \psi(\mathbf{r}, t) \mapsto \{y_i\}, \tag{5}$$

where $\{y_i\}$ is a set of discrete detection positions at the screen. This map may depend on local density structure, neighborhood information, thresholding, and detector geometry.

The key point is that the detector is not modeled as an infinitesimal passive sampler of ρ. It is instead a coarse-grained, finite-resolution operation acting on the propagated field.

OBSERVED BRANCH SELECTION

A central numerical observation is that the detector output may differ qualitatively from the raw Born density. In several runs, the Born density along the detector line exhibits multiple peaks and fine structure, while the actual click events cluster into a much smaller number of narrow horizontal bands.

This suggests the following distinction:

- the wave field remains spatially extended and may pass through multiple available paths,
- but the detector outputs organize into a discrete set of branches.

A useful empirical representation of the click distribution is

$$P_{\text{click}}(y) \approx \sum_k w_k \, \delta(y - y_k), \tag{6}$$

where y_k are branch centers and w_k are empirical branch weights.

The appearance of such discrete branch bands, even when the underlying Born density is continuous and structured, indicates that branch selection can arise from the detector process itself. In this sense, the detector acts as a nontrivial coarse-graining operation rather than a transparent readout of $|\psi|^2$.

BORN PEAKS VERSUS EMERGENT BRANCH CENTERS

A more detailed comparison reveals that the branch centers need not coincide with the maxima of the Born density. Figure 1 shows a representative case from a 400-click simulation. The continuous Born screen density is shown together with the empirical click histogram and the automatically extracted branch centers.

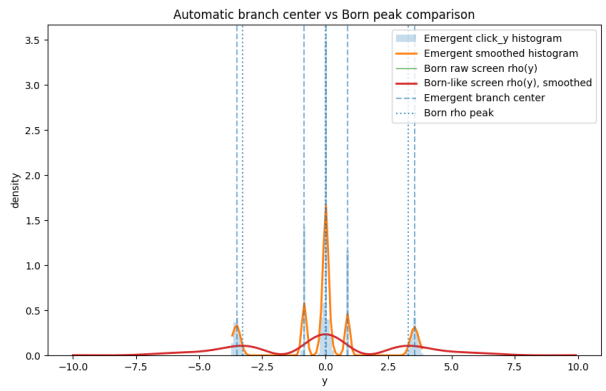

FIG. 1. Comparison between the continuous Born density and the discrete detector output. The red curve shows the smoothed Born density at the screen. The blue histogram and orange curve represent the empirical distribution of detection events.

The comparison in Fig. 1 is important for two reasons. First, it shows that the detector output is much sharper than the underlying Born profile. Second, it shows that one broad Born lobe can correspond to several distinct outcome branches. The detector therefore appears to generate its own effective attractor structure rather than merely reading out preexisting Born maxima.

This observation supports the view that measurement-like selection in the present framework is a structural dynamical process. The result is not simply a continuous density sampled at random, but a small set of stable, discrete outcome channels arising from coarse graining, propagation, and detector geometry.

FIELD-LEVEL SUPERPOSITION VERSUS BRANCH-LEVEL OUTCOME

The numerical results support a distinction between two levels of description.

First, at the field level, the wave function remains extended and may occupy multiple geometrically available paths simultaneously. In this sense, the simulation exhibits the ordinary superpositional character of wave dynamics.

Second, at the branch level, the detector output does not retain the full continuous support of the field. Instead, it selects a discrete subset of stable outcome channels.

This suggests the following qualitative interpretation:

$$\psi \;\longrightarrow\; \rho = |\psi|^2$$

$$\longrightarrow \text{ detector coarse graining } \longrightarrow \text{ discrete branches.} \tag{7}$$

The distinction is important. The field may remain extended across multiple paths, while the observed click

is localized to a single branch. The present framework therefore does not treat the localized outcome as identical to the full wave field itself.

FINITE MEASUREMENT THICKNESS

The most important conceptual proposal of this work is that branch selection may possess a finite spatiotemporal thickness. In other words, the transition from extended superposition to effective branch selection may occur over an interval rather than at a mathematical instant.

We express this schematically by introducing a selection interval

$$t \in [t_1, t_2], \tag{8}$$

during which the propagated field, the geometry, and the detector collectively drive the system toward an effectively discrete outcome structure.

In this picture, measurement is not modeled as a singular event at a single time slice. Instead, it is a gradual process whose effective thickness depends on propagation distance, diffraction regime, and detector structure.

WIDE-SLIT TO DOUBLE-SLIT TRANSITION GEOMETRY

To probe this idea, we propose a numerical experiment consisting of a wide slit followed by a downstream double slit:

$$\text{source} \to \text{wide slit} \to \text{distance } L \to \text{double slit} \to \text{screen}. \tag{9}$$

The distance L between the wide slit and the downstream double slit is the control parameter. This geometry is particularly useful because it naturally interpolates between three regimes:

Near-field regime

For small L, the wave packet has little time to spread after the wide slit. The downstream double slit then behaves approximately like a solid central obstacle with limited effective path separation. In this regime, the field may remain strongly centered and branch selection may already be partially advanced before the second structure becomes effective.

Far-field regime

For large L, the field has time to diffract substantially before reaching the double slit. The second barrier then acts on a broader incoming field, and the usual double-slit interference structure is expected to re-emerge more fully.

Intermediate regime

The most interesting case is intermediate L, where the field is neither strongly localized nor fully spread. In this transition region, branch selection may be partial, unstable, or multi-stage. This is precisely the regime in which a finite measurement thickness should become visible if the selection process is truly gradual.

QUANTITATIVE DIAGNOSTICS

Several numerical diagnostics are useful for characterizing branch selection in this framework.

Branch count

If click events cluster around branch centers y_k, the effective number of branches may be estimated from the number of robust histogram peaks or from cluster extraction methods applied to the click coordinates.

Interference visibility

A standard measure of fringe visibility at the screen is

$$V = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \tag{10}$$

where $I_{\max}$ and $I_{\min}$ are representative neighboring maxima and minima of the screen intensity.

Selection stability

A branch trajectory extracted from local maxima or ridge tracking may be used to quantify how early a branch becomes stable. If a branch coordinate $y_R(t)$ locks into a narrow channel before the detector region is reached, this supports the interpretation that selection develops upstream of the final click event.

Born-to-click deviation

The difference between the raw Born density and the click histogram is itself an important observable. If the detector output is not a direct sampling of the screen density, then

$$P_{\text{click}}(y) \neq C\, \rho_{\text{screen}}(y) \tag{11}$$

for any simple proportionality constant C.

PHASE STRUCTURE AND BRANCH STRUCTURE

Additional diagnostics compare branch structure with phase structure. In the present simulations, branch locations do not appear to follow raw phase stripes directly. Instead, they seem to reflect a combination of amplitude envelope, local geometry, propagation history, and detector coarse graining.

This observation is important because it argues against the oversimplified idea that each branch is merely a direct image of a phase domain. Instead, the detector appears to extract stable outcome bands from the combined wave structure, rather than from phase alone.

RELATION TO OTHER VIEWPOINTS

The present framework is exploratory and sits conceptually between several familiar viewpoints.

Compared with standard textbook measurement, it attempts to model the emergence of a definite outcome as a process rather than taking collapse as primitive.

Compared with Bohmian mechanics, it shares the distinction between an extended wave field and localized effective outcomes, but it does not introduce a full ontological particle trajectory law of Bohmian type.

Compared with decoherence-inspired viewpoints, it emphasizes that a structured detector and propagation geometry may themselves generate effective branch selection and coarse graining.

The present work does not claim equivalence to any of these interpretations. Its aim is more modest: to provide a numerical framework in which finite selection thickness can be studied explicitly.

LIMITATIONS

Several limitations should be emphasized.

- The detector model is phenomenological and not derived from a microscopic detector Hamiltonian.
- Numerical results depend on grid resolution, threshold choices, and branch extraction methods.
- The present framework has not yet been calibrated against laboratory detector response functions.
- The model does not by itself establish a new fundamental interpretation of quantum theory; it only demonstrates that discrete branch-like outputs can emerge in an explicit simulation framework without an inserted collapse term.

CONCLUSION

We have explored a numerical framework in which wave propagation remains continuous, but detector outputs become organized into discrete branches. These observations suggest that measurement-like behavior can emerge from the combination of extended wave dynamics, detector coarse graining, and propagation geometry, even without an explicit collapse operator.

The results motivate the idea of *finite measurement thickness*: branch selection may occur gradually over a finite spatiotemporal region rather than at a single instant. A proposed wide-slit-to-double-slit geometry with variable separation distance provides a direct way to probe this transition numerically.

The comparison between Born maxima and emergent branch centers shows that the detector output is not simply a sharpened copy of the Born distribution. Instead, the detector appears to generate discrete outcome channels with their own effective branch structure.

The framework is exploratory, but it offers a concrete path for studying how definite detector outcomes may emerge from continuous quantum evolution through structured branch selection processes.

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